

3 Partial Derivatives

3.2 Limits and Continuity

Example 3.2.1 Investigate the following limit:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}.$$

What do think should happen as $(x, y) \rightarrow (0, 0)$? What can you find out?

Much as in the case of function of a single variable, we can take limits of functions of two variables. Here is the formal definition:

Definition 3.2.2 If $f(x, y)$ is a function of two variables, and (a, b) is a point in the interior of the domain D of f , then we say that the limit of $f(x, y)$ as (x, y) approaches (a, b) is L if for every number $\epsilon > 0$, there exists a $\delta > 0$ such that whenever

$$\sqrt{(x - a)^2 + (y - b)^2} < \delta,$$

we have that $|f(x, y) - L| < \epsilon$, and we write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L.$$

Example 3.2.3 Show that the following limit does not exist:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}.$$

Theorem 3.2.4 *The properties of the limit which hold for functions of a single variable also hold for functions of two variables:*

1. *The limit of a sum is the sum of the limits.*
2. *The limit of a difference is the difference of the limits.*
3. *The limit of a constant times a function is the constant times the limit of the function.*
4. *The limit of a product is the product of the limits.*
5. *The limit of a quotient is the quotient of the limits, provided that the limit of the denominator is not zero.*

Definition 3.2.5 A function f of two variables is called continuous at (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

We say that f is continuous on D if f is continuous at each point of D .

Example 3.2.6 Find the following limit, if it exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y + 1}{x^3 y^2 - 2x}.$$

Example 3.2.7 Where is the function

$$f(x, y) = \frac{3x^2y}{x^2 + y^2}$$

continuous?

Example 3.2.8 Show that the function

$$f(x, y) = e^{-x^2 - y^2}$$

is continuous everywhere.

Example 3.2.9 If

$$f(x, y) = \frac{xy}{x^2 + y^2},$$

what can be said of the limit

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y)?$$

Example 3.2.10 Determine if the following limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}.$$